Hyeonmin Lee

Assistant Professor
Hanyang University ERICA | Computer Science

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Research Interests

- **Network security**: Securing Internet infrastructure through *data-driven security analysis*, *protocol development and optimization*, and *threat mitigation*.
- **Key Topics**: Domain Name System (DNS), Public Key Infrastructure (PKI), Transport Layer Security (TLS), and Email security.

Education

Ph.D., Computer Science and Engineering, *Seoul National University*, Seoul, South Korea **Mar 2016 — Feb 2022**
- Advisors: Prof. Taekyoung "Ted" Kwon and Prof. Taejoong (Tijay) Chung (co-advisor, *Virginia Tech*)

B.S., Computer Science and Engineering, *Seoul National University*, Seoul, South Korea **Mar 2011 — Feb 2016**

Visiting Student, Information Technology, *Uppsala University*, Uppsala, Sweden **Fall 2014**

Professional Experience

Assistant Professor, Division of Computer Science **Mar 2026 — Present**
Hanyang University ERICA *Ansan, South Korea*

Postdoctoral Research Associate, Department of Computer Science **Oct 2023 — Dec 2025**
University of Virginia *Charlottesville, VA, USA*
- Supervisor: Prof. Yixin Sun

Postdoctoral Researcher, Department of Computer Science and Engineering **Apr 2022 — Sep 2023**
Seoul National University *Seoul, South Korea*
- Supervisor: Prof. Taekyoung "Ted" Kwon

Visiting Student, The Center for Cybersecurity **May 2019 — Aug 2019**
Rochester Institute of Technology *Rochester, NY, USA*
- Supervisor: Prof. Taejoong (Tijay) Chung

Publications (C: Conference | J: Journal | P: Poster)

[C15] DNSSEC Under Interference: A Large-Scale Study of DNS Record Dropping (*to appear*) ***NDSS'27***
Hyeonmin Lee, Shumon Huque, and Yixin Sun Top Conference
In Proceedings of the Network and Distributed System Security Symposium, Seoul, South Korea, Mar 2027

[C14] Revisit DANE for the Web: A Performance Perspective ***IMC'26***
Hyeonmin Lee and Yixin Sun Top Conference
In Proceedings of the 2026 ACM Internet Measurement Conference, Karlsruhe, Germany, Oct 2026
- Artifact: https://github.com/dane-web-study/dane-web-study

[C13] Dissecting NSEC and NSEC3 Deployment in the Wild ***IMC'26***
Yizhe Zhang*, **Hyeonmin Lee***, Shumon Huque, and Yixin Sun Top Conference
In Proceedings of the 2026 ACM Internet Measurement Conference, Karlsruhe, Germany, Oct 2026
* : Equal contribution

[C12] CertDNS: Guaranteeing the Integrity of DNS Records Using PKIX Certificates ***ICC'26***
Hyeonmin Lee, Sangyoon Seok, and Taekyoung "Ted" Kwon
In Proceedings of IEEE International Conference on Communications, Glasgow, Scotland, UK, May 2026

[C11] RFCScope: Detecting Logical Ambiguities in Internet Protocol Specifications ***ASE'25***
Top Conference
Mrigank Pawagi, Lize Shao, **Hyeonmin Lee**, Yixin Sun, and Wenxi Wang
In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering, Seoul, South Korea, Nov 2025
- Artifact: https://github.com/HIPREL-Group/RFCScope/

[C10] Inside Certificate Chains Beyond Public Issuers: Structure and Usage Analysis from a Campus Network ***IMC'25***
Top Conference
Hongying Dong, Yizhe Zhang, **Hyeonmin Lee**, and Yixin Sun
In Proceedings of the 2025 ACM Internet Measurement Conference, Madison, Wisconsin, USA, Oct 2025
- Artifact: https://keyinfra.cs.virginia.edu/cert_chain/

[C9] PQTLS-AD: Post-Quantum TLS Accelerated with DNS ***ICCCN'25***
Sangwon Lim, **Hyeonmin Lee**, Gyeongheon Jeong, and Taekyoung Kwon
In Proceedings of the 34th International Conference on Computer Communications and Networks, Tokyo, Japan, Aug 2025

[C8] PAVE: Information Flow Control for Privacy-preserving Online Data Processing Services ***ASPLOS'25***
Top Conference
Minkyung Park, Jaeseung Choi, **Hyeonmin Lee**, and Taekyoung Kwon
In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Rotterdam, Netherlands, Mar 2025

[C7] Exploring the Ecosystem of DNS HTTPS Resource Records: An End-to-End Perspective ***IMC'24***
Top Conference
Hongying Dong, Yizhe Zhang, **Hyeonmin Lee**, Shumon Huque, and Yixin Sun
In Proceedings of the 2024 ACM Internet Measurement Conference, Madrid, Spain, Nov 2024
- Artifact: https://keyinfra.cs.virginia.edu/dns_http/

[C6] Mutual TLS in Practice: A Deep Dive into Certificate Configurations and Privacy Issues ***IMC'24***
Top Conference
Hongying Dong, Yizhe Zhang, **Hyeonmin Lee**, Kevin Du, Guancheng Tu, and Yixin Sun
In Proceedings of the 2024 ACM Internet Measurement Conference, Madrid, Spain, Nov 2024
- Artifact: https://keyinfra.cs.virginia.edu/mutual_tls/

[C5] DDD: A DNS-based DDoS Defense Scheme Using Puzzles ***ICCCN'24***
Hyeonmin Lee, Taehyun Kang, Sukhun Yang, Jinyong Jun, and Taekyoung Kwon
In Proceedings of the 33rd International Conference on Computer Communications and Networks, Big Island, Hawaii, USA, Jul 2024
- Artifact: https://github.com/Sagit25/DNS-based-DDoS-mitigation

[C4] ZTLS: A DNS-based Approach to Zero Round Trip in TLS handshake ***TheWebConf (WWW)'23***
Top Conference
Sangwon Lim, **Hyeonmin Lee**, Hyunsoo Kim, Hyunwoo Lee, and Ted ''Taekyoung'' Kwon
In Proceedings of the ACM Web Conference 2023 (formerly WWW), Austin, USA, Apr 2023
- Artifact: https://zenodo.org/records/7597964

[C3] Under the Hood of DANE Mismanagement in SMTP ***USENIX Security'22***
Top Conference
Hyeonmin Lee, Md. Ishtiaq Ashiq, Moritz Müller, Roland van Rijswijk-Deij, Taekyoung "Ted" Kwon, and Taejoong Chung
In Proceedings of the 31st USENIX Security Symposium, Boston, USA Aug 2022
- Artifact: https://dane-study.github.io/security2022/security2022-abstract/

[C2] A Longitudinal and Comprehensive Study of the DANE Ecosystem in Email ***USENIX Security'20***
Top Conference
Hyeonmin Lee, Aniketh Gireesh, Roland van Rijswijk-Deij, Taekyoung "Ted" Kwon, and Taejoong Chung
In Proceedings of the 29th USENIX Security Symposium, Boston, USA, Aug 2020
- Artifact: https://dane-study.github.io/security2020/security2020-abstract/

[C1] Development of Cellular Core Network Enabling Network Function Virtualization ***JCCI'18***
Hyeonmin Lee, Junghwan Song, and Taekyoung Kwon
The 28th Joint Conference on Communications and Information, Yeosu, South Korea, May 2018

[J1] TwinPeaks: An Approach for Certificateless Public Key Distribution for the Internet and Internet of Things — ***Computer Networks*** SCIE

Eunsang Cho, Jeongnyeo Kim, Minkyung Park, **Hyeonmin Lee**, Chorom Hamm, Soobin Park, Sungmin Sohn, Minhyeok Kang, and Ted "Taekyoung" Kwon

Elsevier Computer Networks, Jul 2020

[P1] A Longitudinal and Comprehensive Study of the DANE Ecosystem in Email — ***USENIX Security'22***

Hyeonmin Lee, Md. Ishtiaq Ashiq, Moritz Müller, Roland van Rijswijk-Deij, Taekyoung "Ted" Kwon, and Taejoong Chung

Poster Session in the 31st USENIX Security Symposium, Boston, USA, Aug 2022

Grant

A Study for the Future-oriented DANE-based Web Architecture to Solve Problems in the Current TLS-based Web Ecosystem

Primary Investigator — Sep 2022 — Aug 2023

(Funded by *Basic Science Research Program - National Research Foundation of Korea*, ₩60,000,000 ≈ $46,000)

Patents

Homomorphic Cryptographic Parallel Computation Method and Computing Device Performing the Same Method

- Taekyoung Kwon, Minkyung Park, Minhyeok Kang, Selin Chun, **Hyeonmin Lee**
- Registration No. 10-2513552, South Korea, Mar 2023

Method for Performing Mutual Authentication in Communication using Locator ID Separation Protocol, Apparatus, and System for Performing the Same

- Taekyoung Kwon, **Hyeonmin Lee**, Hyunwoo Lee
- Registration No. 10-2476081, South Korea, Dec 2022

Network System and Method for Performing Message Security Thereof

- Taekyoung Kwon, Hyunwoo Lee, Myungchul Kwak, **Hyeonmin Lee**, Junghwan Lim, Yoojung Shin
- Registration No. 10-2265611, South Korea, Jun 2021

Communication Method Based on Integrated Flat ID and System

- Taekyoung Kwon, Hyunwoo Lee, Myungchul Kwak, **Hyeonmin Lee**, Dongjun Lee, Hyunchul Oh
- Registration No. 10-2023115, South Korea, Sep 2019

Talks & Presentations

Invited Talks

- *Why Does DNS Security Remain Vulnerable? A Data-Driven Analysis* — Oct 2026
  National Security Research Institute (NSR), Daejeon, South Korea
- *A New Approach to Web Security Infrastructure: DNS and DANE* — Oct 2026
  2nd BitGaram Energy Valley AI and Information Security Workshop Naju, South Korea

Conference Presentations

- *Revisit DANE for the Web: A Performance Perspective* — Oct 2026
  ACM Internet Measurement Conference (IMC), Karlsruhe, Germany
- *Under the Hood of DANE Mismanagement in SMTP* — Aug 2022
  USENIX Security, Boston, MA, USA
- *A Longitudinal and Comprehensive Study of the DANE Ecosystem in Email* — Aug 2020
  USENIX Security, Online

Technical Writing

- *Under the hood of DANE mismanagement in SMTP* — Sep 2022
  APNIC Blog (https://blog.apnic.net/2022/09/09/under-the-hood-of-dane-mismanagement-in-smtp/)

Professional Services

Program Committee Member

- Technical Program Committee, *ACM Internet Measurement Conference (IMC)* · 2027
- Technical Program Committee, *Network and Distributed System Security Symposium (NDSS)* · 2027
- Technical Program Committee, *IEEE Annual Computer Security Applications Conference (ACSAC)* · 2026
- Technical Program Committee, *Applied Cryptography and Network Security (ACNS)* · 2026
- Poster Program Committee, *ACM Internet Measurement Conference (IMC)* · 2026

Journal Reviewer

- *IEEE Transactions on Dependable and Secure Computing (TDSC)* · 2026
- *Computer Networks* · 2026
- *IEEE/ACM Transactions on Networking (ToN)* · 2024, 2025, 2026
- *IEEE Transactions on Network and Service Management (TNSM)* · 2024, 2025, 2026
- *ACM SIGCOMM Computer Communication Review (CCR)* · 2025
- *Journal of Communications and Networks (JCN)* · 2025
- *IEEE Network* · 2024

External Reviewer

- *Network and Distributed System Security Symposium (NDSS)* · 2025

Advisory Roles

- Technical Advisor, *Korea Internet & Security Agency (KISA)* · 2026

Teaching Experience

(CSE4059) Cloud Security: Fall 2026
(CIS0012) Internet Infrastructure Security: Fall 2026
(GEN1030) Introduction to Computer Systems: Spring 2026

References

Taekyoung “Ted” Kwon

- Professor
- Department of Computer Science and Engineering, Seoul National University, Seoul, South Korea

Taejoong (Tijay) Chung

- Associate Professor
- Department of Computer Science, Virginia Tech, Blacksburg, VA, USA

Yixin Sun

- Assistant Professor
- Department of Computer Science, University of Virginia, Charlottesville, VA, USA